Azentio iMAL AI RFP Copilot

Enterprise Islamic Banking RFP Response & Intelligent Compliance Platform

Hackathon: Azentio GTM AI Hackathon — Challenge B Solution (Level 1 & Level 2)
Target Product Suite: Azentio iMAL Core Banking & Islamic Lending Suite (v15.0)
Team: Synqora | **Architecture**: Java 21 LTS, Spring Boot 3.3.4, PostgreSQL 16 + PgVector 0.8.7 (HNSW)
GitHub Repo: https://github.com/AzentioHackathon-2026/AI-RFP-Response-Agent-synqora.git

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1. Executive Summary & Purpose

Responding to Tier-1 Islamic Banking Request for Proposals (RFPs) is an extraordinarily complex, multi-week endeavor. Tenders regularly feature 100 to 300 technical and regulatory questions spanning Murabaha commodity broker integrations, Ijara asset depreciation schedules, Mudaraba profit distribution smoothing (PER/IRR), immutable AAOIFI Shariah audit trails, and stringent disaster recovery SLAs (RPO < 5 mins, RTO < 15 mins).

The **Azentio iMAL AI RFP Copilot** transforms this manual bottleneck into an autonomous, forensically audited enterprise workflow. Built specifically for Sales Engineers (SEs) and cross-functional bid teams, the platform parses bank tenders from Word (.docx), Excel (.xlsx), CSV, or PDF, performs deep semantic retrieval across Azentio's verified Islamic Banking knowledge vault, classifies requirements into customer-adaptive compliance codes (**S, C, D, T, N**), detects inter-requirement dependencies and multi-product components, distinguishes confirmed product gaps from knowledge gaps, and compiles formatted submission matrices with 100% citation grounding.

2. System Architecture & Tech Stack

The system is built on an enterprise-grade, microservice-ready Spring Boot backend with full vector database integration:

Component	Technology	Specification / Purpose
Backend Framework	Spring Boot 3.3.4 (Java 21 LTS)	REST API, JPA Hibernate ORM, multipart file processing, background daemon executors
Vector Database	PostgreSQL 16 + PgVector 0.8.7	1536-dimensional HNSW cosine index (<=>) for high-speed similarity search across millions of tokens
Semantic Embedding	Deterministic 1536-dim Vectorizer / OpenAI	Built-in offline semantic n-gram vectorizer (100% offline ready) with optional text-embedding-3-small
User Interface	Tailwind CSS + Modular Vanilla JS	Single Page Application (SPA) with 6 dedicated tabs, live KPI meters, and responsive modals
Tender Ingestion	Apache POI, Native DOCX Zip, OpenPDF	Binary multi-format parser for Word (.docx tables), Excel (.xlsx sheets), CSV, and PDF questionnaires
Continuous Learning	JPA Transactional Vector Vault	Approved SE responses dynamically vectorized and added to the past proposal corpus in real time

3. Level 1: Core Grounded Proposal Copilot

Level 1 establishes the bedrock proposal copilot with high-precision grounded synthesis and human-in-the-loop governance:

• **Multi-Format Ingestion & Non-RFP Rejection**: Native parsers extract questionnaires and clause metadata. Non-RFP documents are automatically rejected with defensive validation.

• **Three-Tier Islamic Banking Knowledge Vault**: Segregated index across Product Manuals (iMAL v15.0), Past Scopes of Work (Tier-1 transformations), and Certified Past RFPs.

• **Confidence Scoring & Citation Traceability**: Every drafted sentence includes explicit source file, clause reference, and similarity percentage (0–100%).

• **Quality & Compliance Guardrails**: Detects duplicates, warns on outdated product manuals (e.g. legacy v12/v13 citations), and routes commercial pricing to the Commercial Desk.

• **SE Review Workbench & Continuous Learning Loop**: When a Lead SE clicks 'Approve & Learn', the vetted answer is automatically vectorized into the knowledge base.

4. Level 2: Intelligent Gap Analysis, Compliance & Proposal Assembly

Level 2 extends Level 1 into an autonomous decision-making engine capable of complex requirement analysis, gap categorization, forensic validation, and customer-adaptive assembly:

4.1 Advanced Requirement Interpretation

Detects inter-requirement dependency chains (e.g., REQ-FN-01 Murabaha requires REQ-SEC-01 HSM integration). Maps solution spanning across iMAL Core, Azentio Digital Banking, TradeGate Gateway, and Payment Hub. Flags ambiguous or conflicting SLA requirements.

4.2 Intelligent Compliance Matrix (Adaptive S/C/D/T/N Codes)

Dynamically supports client-defined compliance codes without hardcoding: Code S (Standard Capability), Code C (Configuration Required), Code D (Development Required — mandates engineering man-day estimate), Code T (Third-Party Dependency — mandates partner disclosure), Code N (Not Available / Commercial Confidential). Requires human validation when confidence is below 80%.

4.3 AI-Driven Gap Analysis (Product Gaps vs. Knowledge Gaps)

Crucially separates Confirmed Product Gaps (features requiring code modifications in iMAL) from Knowledge Gaps (supported features lacking bank-specific dark-fiber/latency specs). Classifies into 4 types: Functionality Gap, Integration Dependency, Security Concern, and Missing Documentation. Automatically routes escalation queues to Product Management, Core Engineering, Solution Architecture, and Shariah Board.

4.4 Forensic Response Validation Gate

Gatekeeper that validates answers against source citations before proposal export. Detects Code-to-Narrative mismatches (e.g., marking Code S while the text discusses bespoke engineering sprints) and flags them as BLOCKING errors. Identifies unsupported technological claims and cross-clause contradictions.

4.5 Clarification Query Generation & Prioritization

Generates structured clarification registers (Requirement ID, Ambiguity, Working Assumption, Proposed Clarification). Clusters and deduplicates queries to avoid redundant questions to the client. Prioritizes queries by business impact: Solution Design, Delivery Timeline, or Commercial Scope.

4.6 Proposal Assembly & Readiness Scoring Cockpit

Calculates real-time 0–100% Submission Readiness Score. Enforces customer formatting rules (preserving SEC-X.Y numbering and 250-word answer length caps). One-click export of populated customer response template in .csv (Excel-compatible).

4.7 Collaborative Review Workflow & Selective Regeneration

Assigns clause ownership across 6 department desks. Supports selective section regeneration that strictly preserves already-approved answers.

5. Cross-Department Routing & Escalation Matrix

Requirements requiring escalation or sizing are routed to dedicated department queues in the SE Workbench:

Department Desk	Trigger Criteria & Role	Deliverables / Output
CORE_ENGINEERING	Code D (Development Required), bespoke API connectors, QCB instant payment adapter	Man-day sizing estimates (e.g. 20 man-days), sprint delivery timeline
SOLUTION_ARCHITECTURE	Knowledge Gaps, hybrid disaster recovery, dark-fiber latency, Oracle Data Guard replication	Infrastructure design diagrams, technical clarification queries
PRODUCT_MANAGEMENT	Confirmed Product Gaps, future roadmap commitments, quantum Sukuk features	Roadmap alignment sign-off, release version scheduling (v15.2)
SHARIAH_BOARD	AAOIFI governance, non-halal income purification accounting, automated Murabaha asset sequence	Shariah compliance certificate, FATWA advisory validation note
COMMERCIAL_LEGAL	Code N, pricing matrices, license fees, SLA penalty clauses, third-party broker fees	Bespoke commercial pricing schedule, standard non-disclosure terms
SALES_ENGINEERING	Code S & C standard proposal drafting, customer relationship coordination	Final proposal sign-off, approved RFP response submission pack

6. Complete REST API Reference

The backend exposes complete Level 1 (`/api/v1/rfp`) and Level 2 (`/api/v2/rfp`) endpoints:

Method	Endpoint	Function & Return
POST	/api/v1/rfp/upload	Multipart file upload (.docx, .xlsx, .csv, .pdf) with automatic table parsing
POST	/api/v1/rfp/load-sample-rfp	Instant benchmark loader: ingests Qatar Islamic Bank RFP
GET	/api/v1/rfp/queries?documentId={id}	Returns triaged requirements with confidence scores, citations & status
POST	/api/v1/rfp/responses/{id}/approve	Approves response and dynamically indexes answer into PgVector knowledge vault
POST	/api/v1/rfp/reset	Wipes all active tender records and resets workspace to clean slate
POST	/api/v2/rfp/analyze-level2?documentId={id}	Executes Level 2 engine: S/C/D/T/N coding, gap analysis, and forensic validation audit
GET	/api/v2/rfp/compliance-matrix?documentId={id}	Returns requirement-by-requirement compliance matrix with estimates & disclosures

GET	/api/v2/rfp/gap-analysis?documentId={id}	Distinguishes Product Gaps from Knowledge Gaps with PM/SA/Eng escalation queues
GET	/api/v2/rfp/validation-report?documentId={id}	Forensic audit report (code-to-narrative mismatches, unsupported claims, version alerts)
GET	/api/v2/rfp/clarification-pack?documentId={id}	Clustered and deduplicated clarification pack with impact ratings and assumptions
GET	/api/v2/rfp/submission-checklist?documentId={id}	Real-time submission readiness cockpit (0–100%) and clause compliance checklist
GET	/api/v2/rfp/export/customer-template?documentId={id}	Downloads populated customer response template (.csv) preserving clause numbering
POST	/api/v2/rfp/requirements/{id}/assign	Re-routes requirement ownership to internal desk (PM, Eng, SA, Shariah, Legal)
POST	/api/v2/rfp/regenerate-selective	Regenerates unapproved sections while strictly preserving approved responses

7. Evaluator & Judge Demo Verification Guide

To verify all Level 1 and Level 2 features in the live UI (http://localhost:8080/index.html):

Step 1: Clean Slate — Click the red 'Reset' button in the top navbar to wipe previous test data.

Step 2: Tender Ingest — In Tab 1, browse and select 'Amanah_Crescent_Bank_Islamic_Banking_RFP.docx' and click 'Ingest & Parse' (or click 'Load Demo QIB Tender'). Parses all 33 requirements in seconds.

Step 3: Review Workbench (Tab 2) — Inspect customer response codes (S, C, D, T, N), multi-product spans, dependencies, and citations. Test 'Approve & Learn' on Clause 1 and 'Re-Route Desk' to transfer clause ownership to Core Engineering.

Step 4: Gap Analysis (Tab 3) — View the breakdown distinguishing Confirmed Product Gaps from Knowledge Gaps, with actionable escalation recommendations for Product Management and Engineering.

Step 5: Validation Audit (Tab 4) — Inspect the forensic auditor flagging Code-to-Narrative mismatches and unsupported claims before allowing submission.

Step 6: Clarifications Register (Tab 5) — Inspect clustered queries categorized by Solution Design, Delivery Timeline, and Commercial Scope.

Step 7: Assembly & Matrix (Tab 6) — Review the 0–100% Submission Readiness Cockpit and click 'Download Customer Template (.csv)' to export the finalized proposal.